Chapter 4

# Poynting theorem

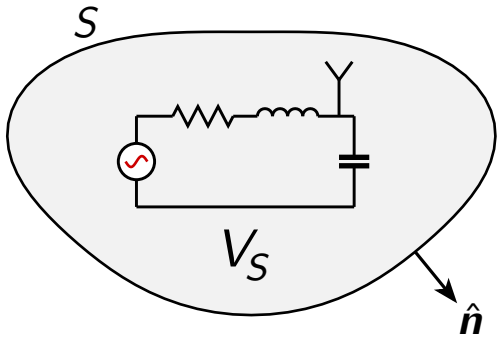
# Poynting theorem - goal

**Question**: how does electromagnetic energy flow in and out of a region $V_S$?

$V_S$ is bounded by the closed surface $S$, with outward normal $\hat{\boldsymbol{n}}$

Start from Faraday-Neumann-Lenz and Ampère-Maxwell (differential form):

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \qquad \nabla \times \boldsymbol{H} = \boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t}$$




**Idea**: turn them into a **power balance**: dot-multiply the first by $\boldsymbol{H}$, the second by $\boldsymbol{E}$

Units check: power densities

$$\boldsymbol{H} \cdot \frac{\partial \boldsymbol{B}}{\partial t} : \mathrm{A/m} \cdot \mathrm{V\ s/m^2} \cdot \mathrm{1/s} = \mathrm{W/m^3} \qquad \boldsymbol{E} \cdot \boldsymbol{J} : \mathrm{V/m} \cdot \mathrm{A/m^2} = \mathrm{W/m^3}$$

# Poynting theorem - dot products

Dot-multiply Faraday-Neumann-Lenz by $\boldsymbol{H}$ and Ampère-Maxwell by $\boldsymbol{E}$:

$$\boldsymbol{H} \cdot \nabla \times \boldsymbol{E} = -\boldsymbol{H} \cdot \frac{\partial \boldsymbol{B}}{\partial t}$$

$$\boldsymbol{E} \cdot \nabla \times \boldsymbol{H} = \boldsymbol{E} \cdot \boldsymbol{J} + \boldsymbol{E} \cdot \frac{\partial \boldsymbol{D}}{\partial t}$$

Subtract the second equation from the first, member by member:

$$\boldsymbol{H} \cdot \nabla \times \boldsymbol{E} - \boldsymbol{E} \cdot \nabla \times \boldsymbol{H} = -\boldsymbol{H} \cdot \frac{\partial \boldsymbol{B}}{\partial t} - \boldsymbol{E} \cdot \boldsymbol{J} - \boldsymbol{E} \cdot \frac{\partial \boldsymbol{D}}{\partial t}$$

# Poynting theorem - vector identity

Vector identity:

$$\nabla \cdot (\boldsymbol{F} \times \boldsymbol{G}) = \boldsymbol{G} \cdot \nabla \times \boldsymbol{F} - \boldsymbol{F} \cdot \nabla \times \boldsymbol{G}$$

With $\boldsymbol{F} = \boldsymbol{E}$ and $\boldsymbol{G} = \boldsymbol{H}$ it is exactly the left-hand side:

$$\boldsymbol{H} \cdot \nabla \times \boldsymbol{E} - \boldsymbol{E} \cdot \nabla \times \boldsymbol{H} = \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H})$$

Local (pointwise) result, with its three terms:

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = - \underbrace{\boldsymbol{H} \cdot \frac{\partial \boldsymbol{B}}{\partial t}}_{\text{magnetic term}} - \underbrace{\boldsymbol{E} \cdot \frac{\partial \boldsymbol{D}}{\partial t}}_{\text{electric term}} - \underbrace{\boldsymbol{E} \cdot \boldsymbol{J}}_{\text{conduction term}}$$

# Poynting theorem - linear isotropic materials

**Assume** linear, isotropic materials, with $\varepsilon$ and $\mu$ constant in time

$$\boldsymbol{D} = \varepsilon \boldsymbol{E}, \quad \boldsymbol{B} = \mu \boldsymbol{H}, \quad \boldsymbol{J} = \sigma(\boldsymbol{E} + \boldsymbol{E}_i)$$

Magnetic term: $$\boldsymbol{H} \cdot \frac{\partial \boldsymbol{B}}{\partial t} = \frac{1}{\mu} \boldsymbol{B} \cdot \frac{\partial \boldsymbol{B}}{\partial t} = \frac{1}{2\mu} \frac{\partial B^2}{\partial t}$$

Electric term: $$\boldsymbol{E} \cdot \frac{\partial \boldsymbol{D}}{\partial t} = \boldsymbol{E} \cdot \frac{\partial (\varepsilon \boldsymbol{E})}{\partial t} = \frac{\varepsilon}{2} \frac{\partial E^2}{\partial t}$$

in teal: $\boldsymbol{a} \cdot \frac{\partial \boldsymbol{a}}{\partial t} = \frac{1}{2} \frac{\partial a^2}{\partial t}$ in red: the constitutive relation used

Conduction term: $$\boldsymbol{E} = \frac{\boldsymbol{J}}{\sigma} - \boldsymbol{E}_i \quad \Rightarrow \quad \boldsymbol{E} \cdot \boldsymbol{J} = \left( \frac{\boldsymbol{J}}{\sigma} - \boldsymbol{E}_i \right) \cdot \boldsymbol{J} = \frac{J^2}{\sigma} - \boldsymbol{E}_i \cdot \boldsymbol{J}$$

# Poynting theorem - local form

Start from

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = -\boldsymbol{H} \cdot \frac{\partial \boldsymbol{B}}{\partial t} - \boldsymbol{E} \cdot \frac{\partial \boldsymbol{D}}{\partial t} - \boldsymbol{E} \cdot \boldsymbol{J}$$

Substitute the three terms:

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = -\frac{1}{2\mu}\frac{\partial B^2}{\partial t} - \frac{\varepsilon}{2}\frac{\partial E^2}{\partial t} - \frac{J^2}{\sigma} + \boldsymbol{E}_i \cdot \boldsymbol{J}$$

Every term is a power density $[\mathrm{W/m^3}]$, valid at each point

# Poynting theorem - integration over $V_S$

Integrate both sides over $V_S$:

$$\int_{V_S} \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) \, \mathrm{d}V = -\int_{V_S} \frac{1}{2\mu} \frac{\partial B^2}{\partial t} \, \mathrm{d}V - \int_{V_S} \frac{\varepsilon}{2} \frac{\partial E^2}{\partial t} \, \mathrm{d}V - \int_{V_S} \frac{J^2}{\sigma} \, \mathrm{d}V + \int_{V_S} \boldsymbol{E}_i \cdot \boldsymbol{J} \, \mathrm{d}V$$

$V_S$ is fixed in time: the time derivative moves outside the integral

$$\int_{V_S} \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) \, \mathrm{d}V = -\frac{\mathrm{d}}{\mathrm{d}t} \int_{V_S} \frac{B^2}{2\mu} \, \mathrm{d}V - \frac{\mathrm{d}}{\mathrm{d}t} \int_{V_S} \frac{\varepsilon E^2}{2} \, \mathrm{d}V - \int_{V_S} \frac{J^2}{\sigma} \, \mathrm{d}V + \int_{V_S} \boldsymbol{E}_i \cdot \boldsymbol{J} \, \mathrm{d}V$$

Divergence theorem on the left-hand side:

$$\int_{V_S} \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) \, \mathrm{d}V = \oint_S (\boldsymbol{E} \times \boldsymbol{H}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$

# Poynting theorem - reordering

Goal: source term on the left, power terms on the right

We found:

$$\oint_S (\boldsymbol{E} \times \boldsymbol{H}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S = -\frac{\mathrm{d}}{\mathrm{d}t} \int_{V_S} \frac{B^2}{2\mu} \, \mathrm{d}V - \frac{\mathrm{d}}{\mathrm{d}t} \int_{V_S} \frac{\varepsilon E^2}{2} \, \mathrm{d}V - \int_{V_S} \frac{J^2}{\sigma} \, \mathrm{d}V + \int_{V_S} \boldsymbol{E}_i \cdot \boldsymbol{J} \, \mathrm{d}V$$

Reordering:

$$\int_{V_S} \boldsymbol{E}_i \cdot \boldsymbol{J} \, \mathrm{d}V = + \int_{V_S} \frac{J^2}{\sigma} \, \mathrm{d}V + \frac{\mathrm{d}}{\mathrm{d}t} \int_{V_S} \frac{B^2}{2\mu} \, \mathrm{d}V + \frac{\mathrm{d}}{\mathrm{d}t} \int_{V_S} \frac{\varepsilon E^2}{2} \, \mathrm{d}V + \oint_S (\boldsymbol{E} \times \boldsymbol{H}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$

# Poynting theorem

$$\int_{V_S} \boldsymbol{E}_i \cdot \boldsymbol{J}\,\mathrm{d}V = \int_{V_S} \frac{J^2}{\sigma}\,\mathrm{d}V + \frac{\mathrm{d}}{\mathrm{d}t}\int_{V_S} \frac{B^2}{2\mu}\,\mathrm{d}V + \frac{\mathrm{d}}{\mathrm{d}t}\int_{V_S} \frac{\varepsilon E^2}{2}\,\mathrm{d}V + \oint_S (\boldsymbol{E}\times\boldsymbol{H})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}S$$

Valid for linear, isotropic materials

Every term is a **power** [W]

Let's read it term by term →

# Poynting theorem - meaning of each term

$$p_g = \int_{V_S} \boldsymbol{E}_i \cdot \boldsymbol{J} \, \mathrm{d}V$$

power of the **generators** (non-electric forces on the charges)

$$p_J = \int_{V_S} \frac{J^2}{\sigma} \, \mathrm{d}V$$

power **dissipated** by Joule effect (collisions)

$$E_m = \int_{V_S} \frac{B^2}{2\mu} \, \mathrm{d}V$$

**magnetic** energy stored in $V_S$

$$E_e = \int_{V_S} \frac{\varepsilon E^2}{2} \, \mathrm{d}V$$

**electric** energy stored in $V_S$

$$p_P = \oint_S (\boldsymbol{E} \times \boldsymbol{H}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$

power **leaving** $V_S$ through $S$

$\boldsymbol{S} = \boldsymbol{E} \times \boldsymbol{H}$ is the **Poynting vector** $[\mathrm{W/m^2}]$

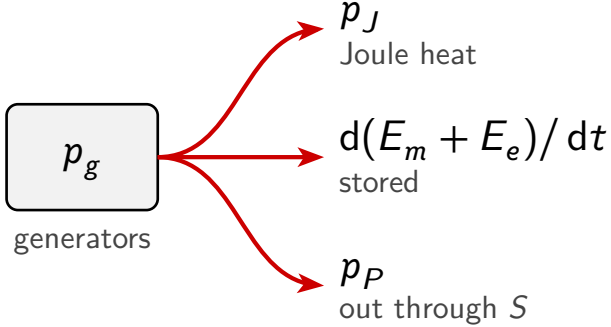
# Poynting theorem - energy balance

The Poynting theorem is an **energy balance** for $V_S$

$$p_g = p_J + \frac{\mathrm{d}}{\mathrm{d}t}(E_m + E_e) + p_P$$

The power produced by the generators is

- partly **dissipated** by Joule effect
- partly **stored** as magnetic and electric energy
- partly **transported out** of $V_S$ through $S$




Next: we use it to prove that **any** e.m. problem has a **unique** solution